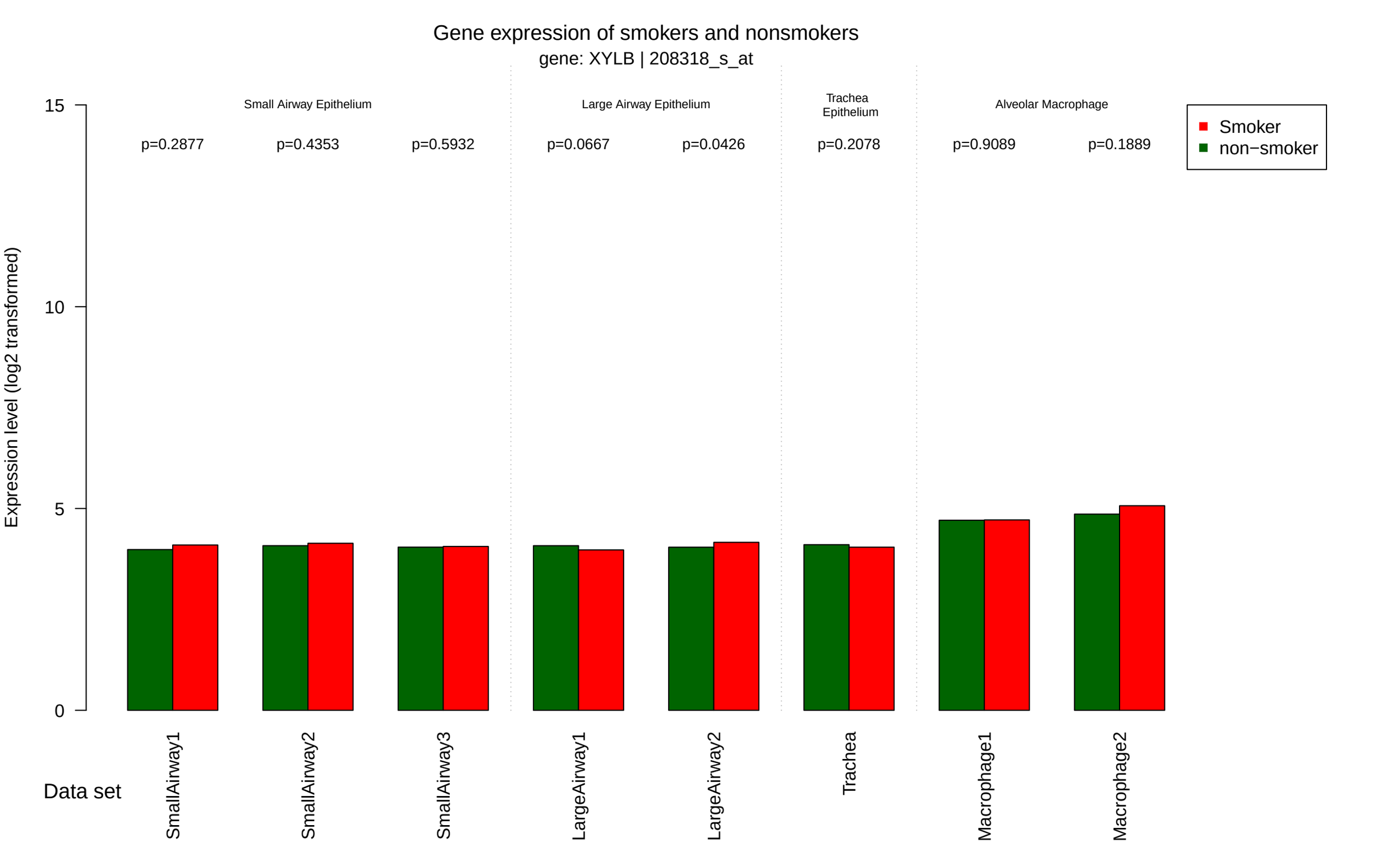
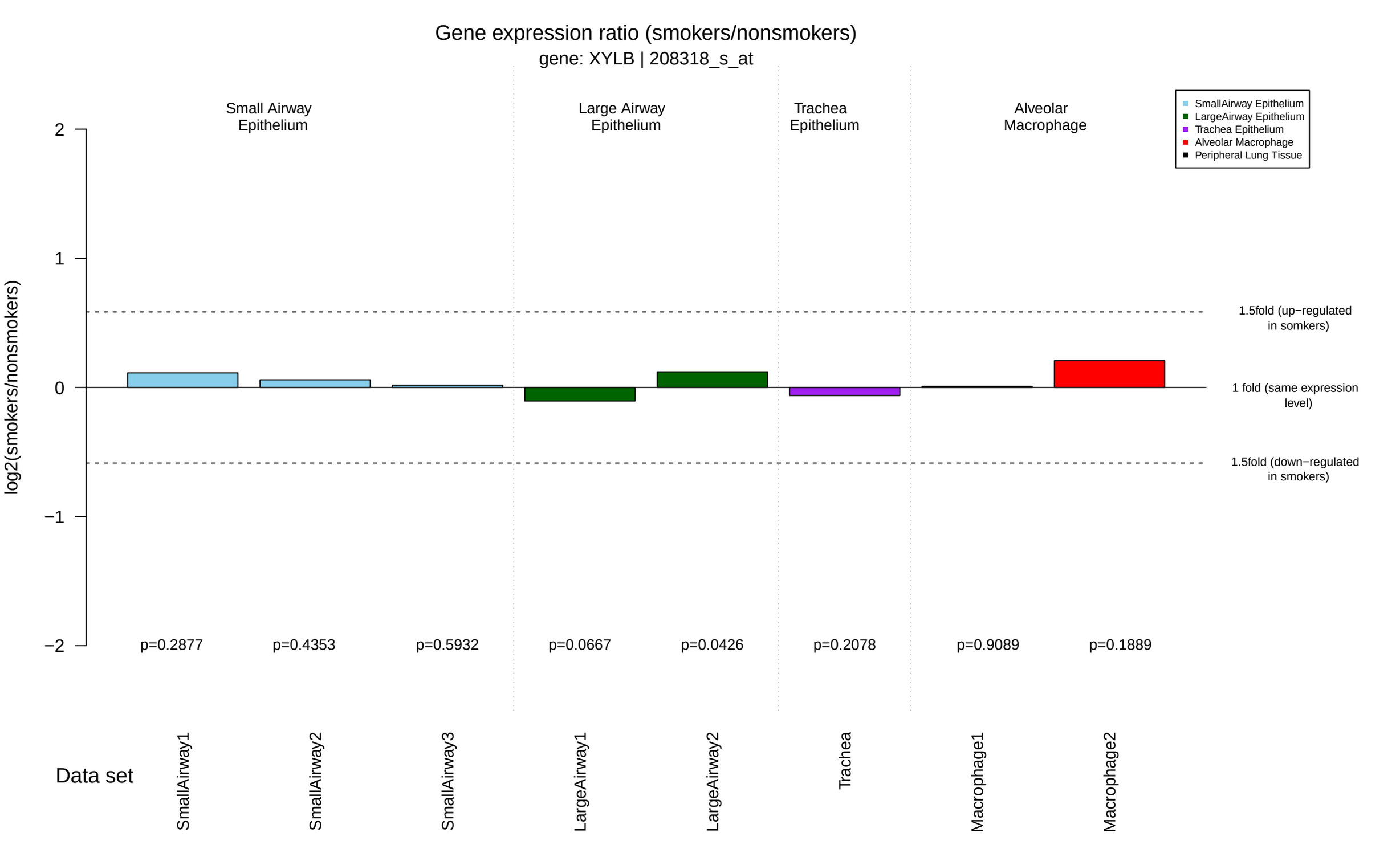
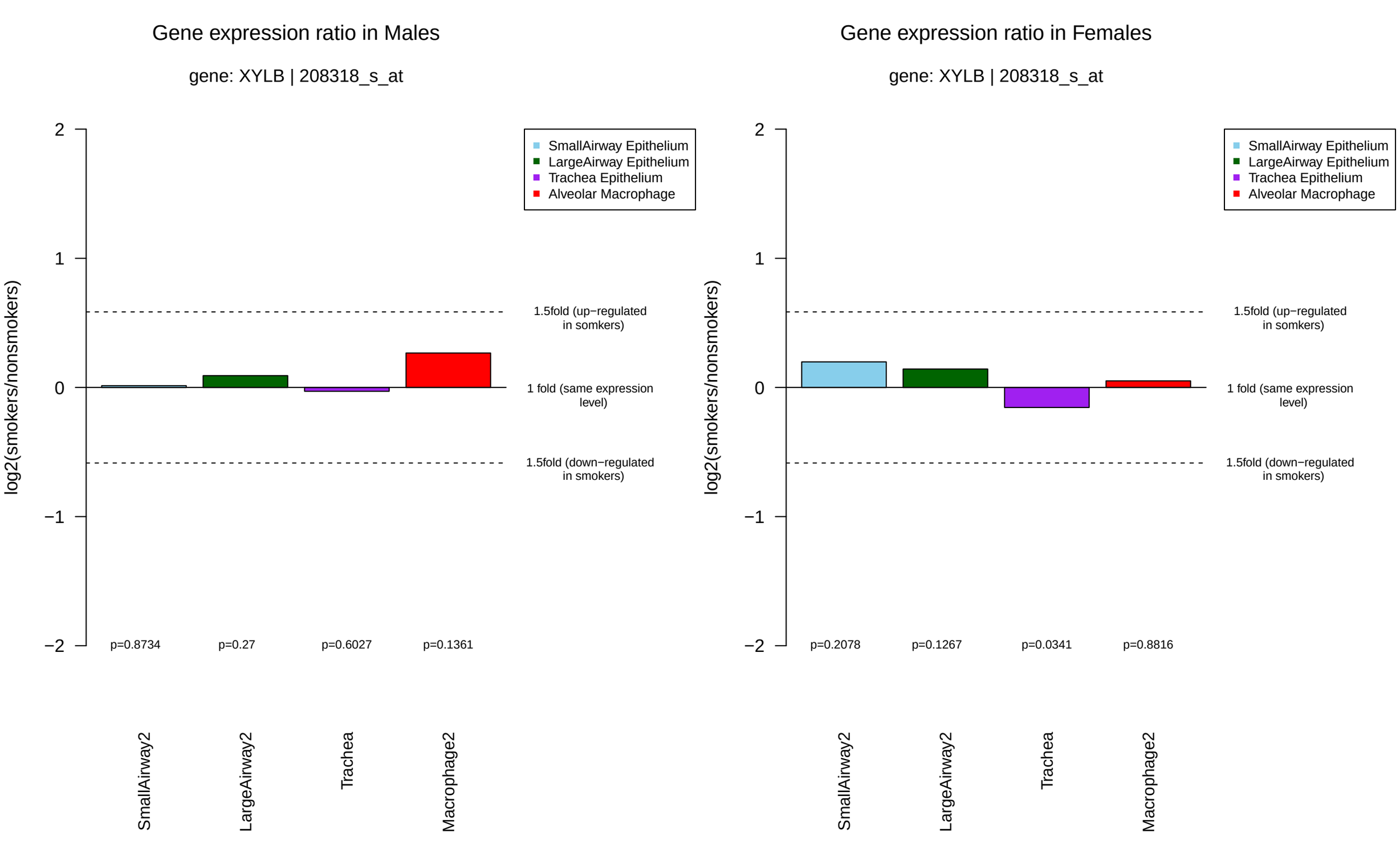
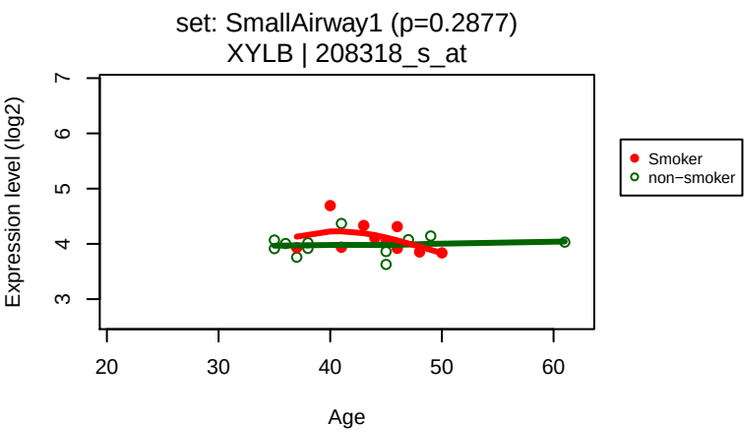
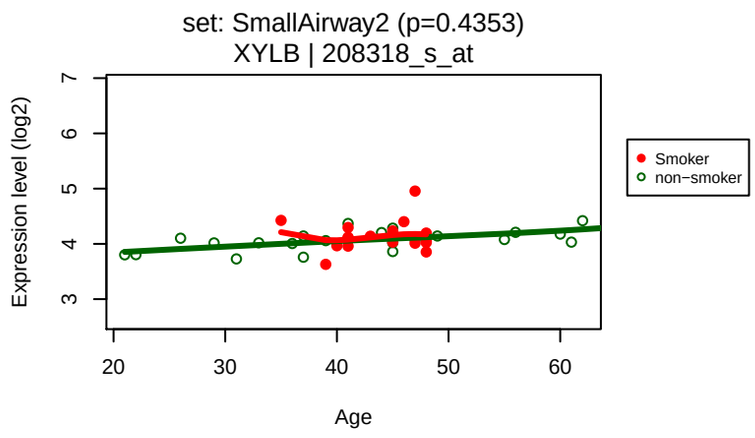
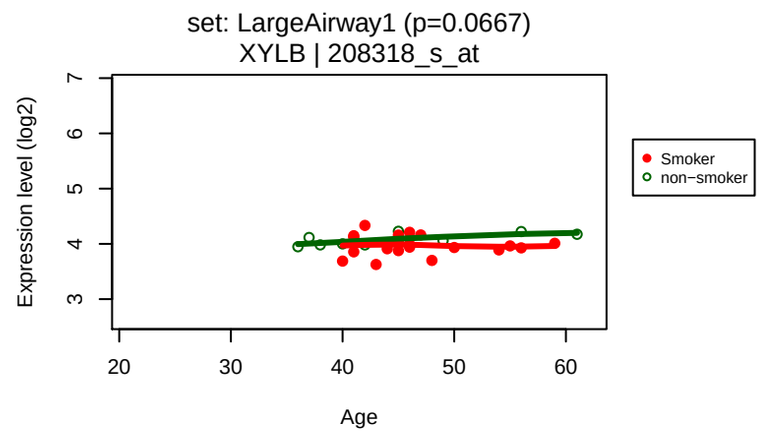

Gene expression of smokers and nonsmokers
gene: XYLB | 208318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2877
p=0.4353
p=0.5932
p=0.0667
p=0.0426
p=0.2078
p=0.9089
p=0.1889
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

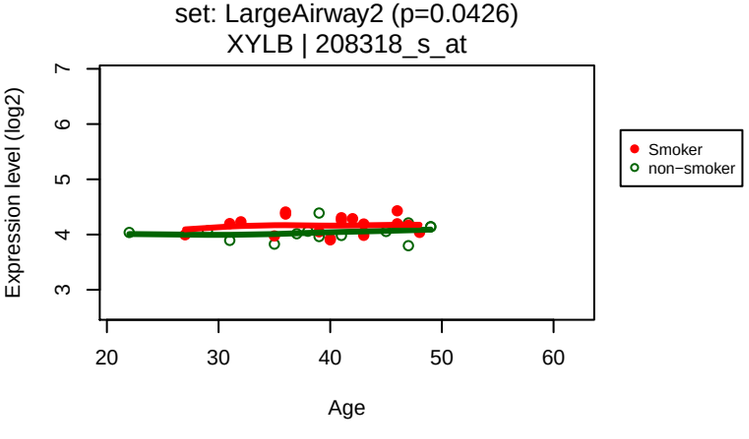

Gene expression ratio (smokers/nonsmokers)
gene: XYLB | 208318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2877
p=0.4353
p=0.5932
p=0.0667
p=0.0426
p=0.2078
p=0.9089
p=0.1889
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

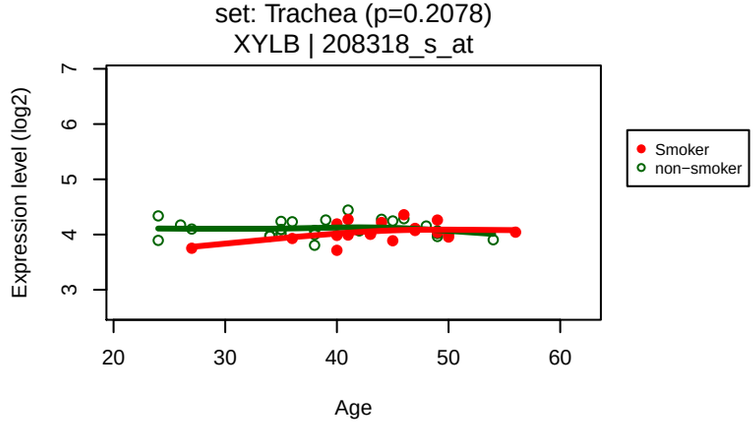

Gene expression ratio in Males
gene: XYLB | 208318_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8734
p=0.27
p=0.6027
p=0.1361
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XYLB | 208318_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2078
p=0.1267
p=0.0341
p=0.8816
SmallAirway2
LargeAirway2
Trachea
Macrophage2

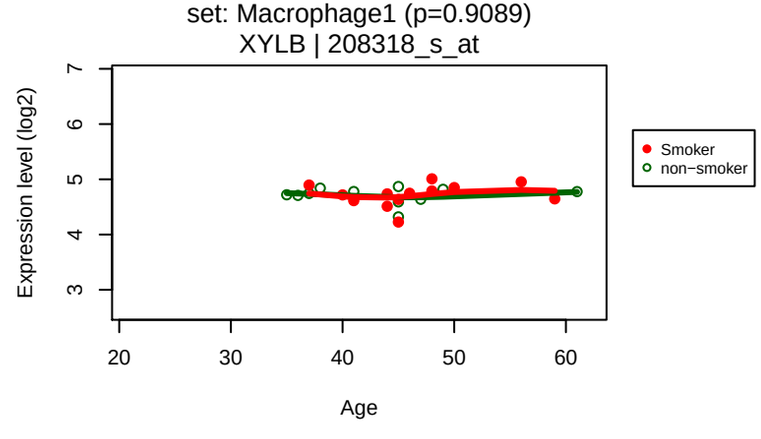

set: SmallAirway1 (p=0.2877)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker

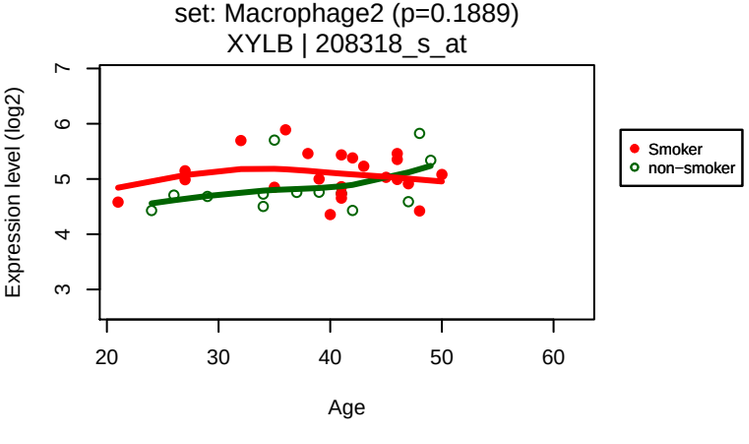

set: SmallAirway2 (p=0.4353)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker

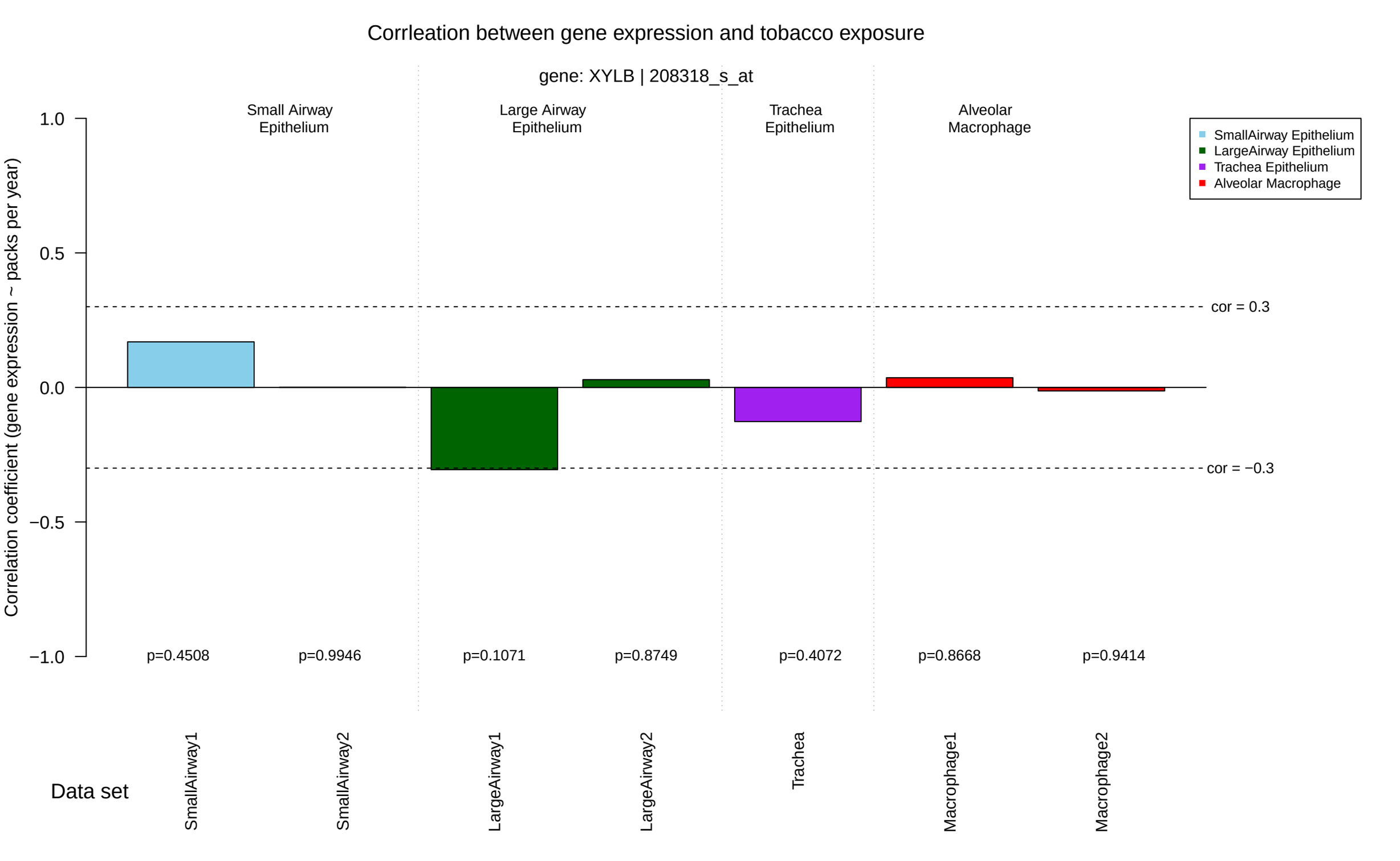

set: LargeAirway1 (p=0.0667)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0426)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2078)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9089)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1889)
XYLB | 208318_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XYLB | 208318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4508
p=0.9946
p=0.1071
p=0.8749
p=0.4072
p=0.8668
p=0.9414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2